Megha Project Titles 2025-2026

SL.NO	FACULTY NAME	PROJECT TITLES
1.	Dr. Abhinandan P. Shirahatti	**Problem 1:** **AI-Driven Remote Monitoring and Early Warning System for Heart Attack Risk in the Indian Population** **Objectives:** 1. To design a wearable device that captures cardiac vitals continuously (ECG, HRV, BP). 2. To develop AI/ML models capable of predicting early warning signs of myocardial infarction. 3. To integrate real-time alerts through mobile/cloud platforms for patients and emergency services and pilot-test the system among at-risk individuals across urban and semi-urban Indian populations. **Problem 2:** AI-Based Mental Health Screening for Adolescents in Indian Schools **Objectives:** 1. To develop a mobile-based AI chatbot that screens for early signs of anxiety, depression, and stress among school students. 2. To integrate voice sentiment and text analysis for emotion detection and mental wellness scoring. 3. To validate the tool through clinical psychologist collaboration in selected schools. **Problem 3:** Predictive Analytics for Maternal Health Risk in Remote Rural Areas **Objectives:** 1. To design a mobile-based application for recording maternal health indicators (anemia, BP, fetal movements). 2. To create a machine learning model to predict high-risk pregnancy conditions (e.g., preeclampsia, gestational diabetes). 3. To train ASHA workers and ANMs to collect data and provide basic alerts and integrate the system with the local PHC for timely referrals and antenatal care improvements.

2.	Mrs. Bhagyashree Udagave	**Problem 1:** Tax evasion significantly impacts a nation's economy by reducing government revenue and increasing the burden on honest taxpayers. Detecting potential tax evasion requires analysing large volumes of financial and transactional data to identify suspicious patterns. The goal is to develop a system that can assist tax authorities by flagging high-risk entities or individuals using data-driven techniques. **Objectives:** 1. Data Collection and Preprocessing(Gather and preprocess datasets related to income, transactions, property ownership, bank records, and tax filings to prepare the data for analysis) 2. Tax Evasion Risk Analysis using Machine Learning. 3. Design a user-friendly dashboard for tax authorities to visualize risk levels, flag suspicious activities, and generate reports. **Problem 2:** Developing technology-such as software or devices-capable of accurately converting spoken language to Indian Sign Language (ISL). **Objectives:** 1. To build a system that converts spoken language into Indian Sign Language (ISL) gestures 2. To create a mobile or display-based app that shows ISL signs for public announcements 3. To develop a mobile tool using a camera that helps hearing people understand the ISL signs used by deaf individuals, making communication between them easier. **Problem 3:** A comprehensive Alumni Association platform **Objectives:** 1. To design and develop a scalable and secure Alumni Association platform accessible via both web and mobile applications 2. To implement core functionalities such as donation management, job and networking portals, event registrations, and alumni directory searches 3. To integrate success story tracking, feedback mechanisms, and personalized user experiences that promote knowledge sharing, mentorship

3.	Mr. Chaitanya B. Pednekar	**Problem 1:** **Developing Robust Multilingual Sentiment Analysis Models for Code-Mixed Social Media Texts** **Objectives:** 1. To design and train deep learning models capable of handling syntactic and semantic irregularities in code-mixed datasets. 2. To curate and annotate large-scale social media datasets in code-mixed languages (e.g., Hindi-English, Marathi-English) 3. To evaluate cross-lingual transfer learning techniques for improving sentiment prediction accuracy across various code-mixed language pairs. **Problem 2:** **Incorporating Contextual and Cultural Semantics in Code-Mixed Sentiment Analysis** **Objectives:** 1. To explore the impact of cultural context and idiomatic expressions in sentiment orientation of code-mixed texts. 2. To integrate external knowledge bases and cultural embedding into sentiment classification models. 3. To develop context-aware evaluation metrics that reflect sentiment polarity shifts due to cultural nuances in code-mixed language. **Problem 3:** **Leveraging Multimodal Cues for Sentiment Analysis in Code-Mixed Content** **Objectives:** 1. To design multimodal neural architectures that effectively learn sentiment cues from diverse input formats. 2. To design multimodal neural architectures that effectively learn sentiment cues from diverse input formats. 3. To measure the improvement in sentiment classification accuracy by incorporating non-textual sentiment indicators

4.	**Dr Mamata S Kalas**	**Problem 1:** **Implementation of Self-Supervised Learning (SSL) for Time Series Analysis Analysis** **Objectives:** 1. To review existing surveys related to SSL and time series and then provide a new taxonomy of existing time series SSL methods. 2. To analyse SSL properties in time series modelling tasks, and give reliable benchmark evaluation 3. To facilitate the experiments and validation of time series SSL methods **Problem 2:** **Enhancing Autonomous Vehicle Navigation Using Deep Learning-Based Scene Understanding.** **Objectives:** 1. To review AV navigation Techniques for scene understanding 2. Develop Deep Q-Learning reinforcement learning system for real-time navigation and vehicle control situations 3. To Analyse the Results and Performa comparative AV Navigation analysis **Problem 3:** **Design Vision-Based Emotion Interpretation Network for Emotional Video Captioning** **Objectives:** 1. To model the emotion distribution over an open psychological vocabulary to predict the emotional state of videos. 2. To incorporate visual context, textual context, and visual-textual relevance into an aggregated multimodal contextual vector to enhance video captioning 3. To optimize the network in a new emotion-fact for analysing Emotional Indication Loss and Factual Contrastive Loss

5.	**Mrs. Suchita Sagar Patil**	**Problem 1:** **LLM-Powered Sentiment Analysis for Product Reviews** **Objectives:** **1. To develop and evaluate a sentiment analysis model using Large Language Models (LLMs) that accurately classifies product reviews into categories such as positive, negative, or neutral.** **2. To enhance understanding of consumer opinions by leveraging the contextual and semantic capabilities of LLMs, thereby improving the interpretability and depth of sentiment analysis beyond traditional methods.** **3. To compare the performance of LLM-based sentiment analysis with conventional machine learning and deep learning techniques in terms of accuracy, robustness, and adaptability across diverse product categories and review platforms.** **Problem 2:** **Sarcasm Detection in Social Media with Images** **Objectives:** **1. To develop a multi-modal sarcasm detection model that integrates both textual content and visual cues (e.g., memes, emojis, images) from social media posts for improved accuracy.** **2. To analyze the correlation between visual elements and sarcastic intent in social media content by extracting features such as facial expressions, gestures, and symbolic context in images.** **3. To create a labeled dataset of multi-modal sarcastic and non-sarcastic posts from platforms like Twitter, Instagram, and Reddit, supporting supervised learning and benchmarking of detection models.** **4.** **Problem 3:** LLM-Enhanced Customer Feedback Analysis with videos **Objectives:** **1. Leverage Large Language Models (LLMs) to analyse textual and spoken feedback.** **2. Integrate video content understanding with textual analysis** **3. Develop an end-to-end pipeline for automated feedback interpretation and reporting**

6	Mr. Sameer Patil	**Problem 1:** **Algorithmic Study of Financial Market** **Objectives:** 1. **Identification of Data Sources & Validity of sources** 2. **identification of limitations of existing approaches & system** 3. **defining solution scope & implementation of same** **Problem 2:** **Optimization Algorithms of ML** **Objectives:** 1. **Study of optimization algorithms and hyper parameter tuning techniques** 2. **Implementation & comparative study of these algorithms** 3. **Result analysis and conclusion - publication in paper** **Problem 3:** **Object detection & summary generation** **Objectives:** 1. **Design the system which can detect the object - in the given image** 2. **generate summary text about the object** 3. **answer - interact with the user based on the entered object.**
7.	Mr. MAHESH S. SALUNKHE	**Problem 1:** **Plant disease detection using cycleGAN** **Objectives:** 1. **To build a Dataset** 2. **To apply cycleGAN for detecting similar diseases in different plants** 3. To develop a mobile app/web app **Problem 2:** **To find new design patterns for textile fabrics** **Objectives:** 1. **To build a dataset of textile design patterns** 2. **To design deep learning models to find new design patterns from the standard patterns** 3. To develop a mobile app **Problem 3: Finding details of textile fabrics** **Objectives:** 1. **To build a textile fabrics dataset** 2. **To design an efficient deep learning model to find fabric details** 3. **To apply GAN models to generate new fabric patterns**

8.	**Mrs. Vaishali G Powar**	**Problem 1:** **Smart Resource Management for common facilities in residential apartment** **Objectives:** 1. To identify and analyse the current inefficiencies in the usage of common facilities in residential apartments. 2. To design a prototype smart monitoring system using IoT and real-time data collection for tracking resource consumption and facility usage. 3. To implement and evaluate intelligent decision-making algorithms for resource scheduling and billing, predictive maintenance, and energy-saving recommendations based on collected data. **Problem 2:** **Smart Software Management Across a Local Area Network** **Objectives:** 1. To identify and analyse the current inefficiencies in the software management over local area networks. 2. To design a smart smart software management system for local area network. 3. To implement centralized software management integrated environment. **Problem 3:** **Lightweight Straggler Detection in Heterogeneous Distributed Systems** **Objectives:** 1. To study the impact of hardware heterogeneity on node performance and straggler identification. 2. To develop a lightweight profiling system that considers both historical performance and workload type. 3. To validate the approach in a simulated heterogeneous environment and measure improvements in task scheduling efficiency.
9.	**Mr. Sanjay S .Patil**	**Problem 1:** **Artificial Intelligence Chatbot for Customer Support** **Objectives:** 1. Monitor customer service inquiries. 2. Respond to customer inquiries efficiently 3. To handle complex support requests **Problem 2:** **IoT-based Smart Home Automation System** **Objectives:** To control and monitor their home environments remotely To achieve energy efficiency.

		Optimize energy consumption and reduce costs **Problem 3:** Advanced Healthcare Information Management System **Objectives:** 1. To achieve streamline operations in healthcare facilities 2. To enables secure access to patient data 3. To integrate telehealth services.
10.	Dr. Lingaraj A. Hadimani	**Problem 1:** AI-Powered Campus Navigator for Visually Impaired **Objectives:** 1. To design a real-time object detection and obstacle alert system using a mobile camera and deep learning models. 2. To integrate GPS and Bluetooth beacon-based indoor-outdoor navigation for seamless campus guidance. 3. To develop a voice-command interface for two-way interaction, feedback, and safety alerts. **Problem 2:** Smart Traffic Violation Detection using CCTV and Computer Vision **Objectives:** 1. To train image recognition models to detect specific violations using CCTV footage in real-time. 2. To integrate number plate recognition (ANPR) for automated challan generation. 3. To provide a live dashboard for authorities to monitor violation trends and identify hotspots. **Problem 3:** Personalized Mental Health Chatbot with Emotion Detection **Objectives:** 1. To implement sentiment and emotion analysis from user messages using NLP models (e.g., BERT, LSTM). 2. To provide context-aware responses and tips based on cognitive behavioral therapy principles. 3. To ensure user privacy, session logs, and emergency alerts in cases of detected distress signals.
11.	Mr. Vivek Ravindra Ugale	**Problem 1:**

		Educational Gamification Using Interactive Game Development **Objectives:** 1. Leverage game-based elements to transform traditional learning environments, encouraging students to take an active role in their education and retain information better through the fun, immersive experiences of games.To integrate GPS and Bluetooth beacon-based indoor-outdoor 2. Use game-based challenges to promote higher-order thinking (like analysis, synthesis, and evaluation) instead of passive content consumption. This helps students apply their knowledge interactively rather than just memorizing facts. 3. Create interactive educational games that adapt to different learning paces, levels of proficiency, and preferred learning styles (visual, auditory, kinesthetic) to provide personalized learning experiences. **Problem 2:** Developing an E-commerce Website Utilizing 3D Models Through Full Stack Web Development **Objectives:** 1. Implement 3D models of products that users can interact with, rotate, zoom, and explore in detail on the e-commerce platform. 2. Develop a full-stack web application that connects the front-end (user interface) with the back-end (database and server) to ensure smooth product browsing, purchasing, and transaction processing. 3. Ensure the 3D models are optimized for performance across a wide range of devices (desktops, tablets, and smartphones) and that they load efficiently without compromising the site's performance. **Problem 3:** Designing a Farmer Robot for Weed Detection and Removal Using Computer Vision and Object Detection **Objectives:** 1. Design and build a robot equipped with sensors and actuators that can autonomously navigate through fields to identify and remove unwanted weeds. 2. Use computer vision algorithms (like Convolutional Neural Networks) and object detection techniques (e.g., YOLO, Faster R-CNN) to distinguish weeds from crops and other objects. 3. Develop a real-time control system that integrates the weed detection algorithm with robotic movement and action mechanisms (such as cutting, pulling, or spraying).

12	Mrs. Anuradha Sangram Solanki	**Problem 1:** **VR learning app using generative AI** **Objectives:** 1. **VR Learning Environment Design and development of the virtual reality scenes.** 2. **Generative AI Content Engine Use generative AI (like GPT or other models) to create learning content** 3. User Interaction & Performance Tracking **Problem 2:** **AR Quiz Game with Generative AI(AI generates new questions. Questions or objects appear in AR on a table or book surface)** **Objectives:** 1. **Augmented Reality Interface (Develop the AR environment using tools like Unity with Vuforia or ARCore/ARKit)** 2. **Generative AI Question Engine** 3. User Interaction & Performance Tracking (Track user's answers, time taken, and accuracy) **Problem 3:** VR-Based Medical Counselling App Using AI **Objectives:** 1. **VR Environment for Medical Counselling** 2. **AI-Powered Virtual Counsellor** 3. **User feedback and analysis**
13	Mr. Ajit Bhimrao Patil	**Problem 1:** **Image recognition for Medical Diagnosis** **Objectives:** 1. **Enhance Diagnostic Accuracy and Reduce Human Error** 2. **Improve Efficiency and Workflow Optimization** 3. Facilitate Personalized and Predictive Medicine **Problem 2:** **soybean leaf disease detection** **Objectives:** 1. **Develop a Robust and Accurate Image Recognition Model for Disease Classification:** 2. **Enable Early and Rapid Disease Detection for Timely Intervention:** 3. **Provide an Accessible and User-Friendly Tool for Farmers** **Problem 3: Automated Daily Operations Management System for Petrol Pumps in Rural Maharashtra** **Objectives:** 1. **Automate and Centralize Daily Operational Data Management:** 2. **Enhance Inventory Control and Financial Accountability** 3. **Improve Operational Efficiency and Staff Management patterns**

14	**Mr. Arun Desai**	**Problem 1:** **Build an autonomous agent that extracts structured job data from unstructured job postings (websites, PDFs, emails).** **Objectives:** **1. Use GenAI to extract key job fields — title, skills, experience, location, salary, etc. — from semi-structured or unstructured content.** 2. **Deploy an agent to crawl and monitor job boards or company career pages periodically.** **3. Store extracted data into a structured format (CSV, JSON, database) and allow query-based interaction ("List jobs requiring Python and remote work").** **Problem 2:** **Skill Gap Analyzer for Job Seekers** **Objectives:** 1. **Extract required skills and qualifications from job descriptions using GenAI.** **2. Parse candidate resumes and match them semantically with job requirements.** **3. Generate a personalized skill-gap report with suggested learning resources, using an agentic assistant.** **Problem 3: Job-to-Career Path Generator** **Objectives:** **1. Extract job titles, required skills, and growth indicators (seniority level, industry trends) from job listings using GenAI.** **2. Use an agentic loop to trace typical career paths based on historical job data or knowledge graphs (e.g., Junior Developer → Developer → Team Lead → Engineering Manager).** **3. Present personalized career path suggestions to users with missing skill recommendations and role transitions.**
15	**Mr. Samadhan Shivaji Palkar**	**Problem 1:** **Secure Federated Learning over Blockchain-enabled 6G Networks** **Objectives:** **1. Design a permissioned blockchain framework to record and verify federated learning updates securely across edge devices using smart contracts** 2. **Develop an optimized encryption or differential privacy mechanism to ensure that local model updates remain confidential during transmission over 6G.** **3. Implement and evaluate the system on a testbed (e.g., using Raspberry Pi clusters with 6G simulators)**

		and publish results comparing latency, throughput, and security with traditional FL approaches. **Problem 2:** **Intrusion Detection System for 6G-enabled IoT using Blockchain and Machine Learning** **Objectives:** 1. Develop a machine learning model (e.g., hybrid deep learning) that can detect anomalies in real-time 6G IoT traffic with high accuracy and low false positives 2. Design a blockchain mechanism for decentralized sharing and validation of detected threat signatures among distributed IDS nodes. 3. Deploy the system in a simulated IoT-6G environment (e.g., NS-3/OMNeT++ plus Hyperledger Fabric) and benchmark its performance against conventional IDS solutions. **Problem 3:** Privacy-preserving Smart Contract Framework for 6G Smart Cities **Objectives:** 1. Design smart contracts that allow users to control access to their data using privacy-preserving techniques (e.g., homomorphic encryption, zero-knowledge proofs). 2. Integrate a machine learning component that monitors smart contract interactions for abnormal or malicious behaviour and adapts policies dynamically. 3. Prototype the framework using a blockchain platform (like Ethereum or Hyperledger) with a 6G network simulator and validate it through realistic smart city scenarios (e.g., connected vehicles, e-health).
16	**Dr. Deepali Kishor Jadhav**	**Problem 1:** **Stock Market Trend Prediction using Machine Learning with a Multimodal Approach** **Objectives:** 1. Collect historical stock data, financial news, and social media sentiment data for selected stocks 2. Engineer features using technical indicators (MACD, RSI), and extract sentiment scores from text using NLP models. 3. Design and implement a multimodal ML model using early or late fusion techniques (e.g., LSTM + BERT, Random Forest + Sentiment Vector). Train and evaluate the model using classification metrics (accuracy, precision, recall, F1-score) and financial metrics (e.g., profit/loss simulation). Compare performance of multimodal model with uni-modal baselines (price-only or text-only). **Problem 2:** **Brain Tumor Multiclass Classification from MRI Images using Deep Learning with a Multimodal Approach**

		Objectives: 1. Collect and preprocess MRI images and associated patient metadata or clinical notes (synthetic or real) 2. Design and train a CNN-based model (e.g., VGG16, ResNet, or custom CNN) for image-based classification. Apply NLP techniques (e.g., BERT, BioBERT) to extract semantic embeddings from clinical notes or symptom descriptions. 3. Compare performance with uni-modal models (image-only and text-only). **Problem 3:** Early Prediction of Student Academic Performance using Machine Learning Techniques **Objectives:** 1. Collect and preprocess academic and behavioral data of students. 2. Build predictive models using algorithms like Decision Trees, Random Forest, SVM, and Neural Networks. 3. Compare model performance using metrics like accuracy, precision, recall, and F1-score.
17	**Dr. Sonal Ayare**	**Problem 1:** Create an Annual Report Portal for institute where all the departmental reports can be integrated and customized **Objectives:** 1. Develop a centralized, secure, and user-friendly Annual Report Portal to enable all departments of the institute to upload, manage, integrate, and customize their reports. 2. The portal should facilitate seamless collaboration, real-time updates, and generate consolidated annual reports in multiple formats. 3. Develop a centralized, secure, and user-friendly Annual Report Portal **Problem 2:** Automated Bus Scheduling and Route Management System **Objectives:** 1. To design and implement an Automated Bus Scheduling and Route Management 2. Route Management System that optimizes bus fleet operations by generating efficient schedules and dynamic routes based on real-time data 3. The system aims to improve punctuality, reduce operational costs, and enhance passenger convenience **Problem 3:** Software solutions to identify users behind Telegram, WhatsApp and Instagram based drug trafficking. **Objectives:**

		1. User Identification & Attribution, Communication Pattern Analysis 2. Real-time Monitoring & Alerting 3. Data Privacy & Legal Compliance
18	Shubhada Sawakhande	**Problem 1:** Smart Irrigation System for Precision Farming **Objectives:** 1. To develop smart irrigation system for farmers. 2. To ease their work. 3. step towards digitisation **Problem 2:** Developing innovative solution for efficient management of waste and conversion to valuable products, Waste to Energy-waste recycling **Objectives:** 1. To develop smart waste management system 2. To take a step towards digitisation 3. To make it easier for smart waste management **Problem 3:** Software solutions to identify users behind Telegram, WhatsApp and Instagram based drug trafficking. **Objectives:** 1. Smart traffic management system Real-time Monitoring & Alerting 2. Effective AI Based solution 3. To execute management centrally
19	Reecha Suryavanshi	**Problem 1:** Use of Digital Technology to calculate Water Footprints for different Agricultural Products 1. To develop a digital framework that collects and processes data related to crop type, irrigation methods, climate, and soil characteristics to calculate the water footprint of agricultural products with high accuracy. 2. To integrate remote sensing, GIS, or IoT technologies (such as soil moisture sensors or satellite imagery) to enable real-time or location-specific monitoring of water usage in farming practices. 3. To generate comparative water footprint reports for different crops and regions, providing actionable insights for farmers, policymakers, and agribusinesses to adopt more sustainable and water-efficient agricultural practices. **Problem 2:** Smart Competency Diagnostic and Candidate Profile Score Calculator

		Objectives: 1. To design an intelligent assessment system that identifies and evaluates candidates' core competencies — including technical, analytical, communication, and behavioral skills — through customizable digital tests or input modules. 2. To develop a scoring algorithm that calculates a comprehensive candidate profile score by analyzing assessment results, academic background, experience, and other relevant parameters using data-driven techniques. 3. To present the candidate profile in a structured format with detailed feedback and recommendations, helping recruiters, educators, or mentors make informed decisions for hiring, training, or skill development. **Problem 3:** Road Transport Network Telematics Develop a telematics solution to enable efficient trucking operations for the long haul to connect the country through route optimization, live tracking and monitoring, optimal capacity utilization analysis and to enable appropriate response **Objectives:** 4. To design a telematics-based system that enables real-time tracking, monitoring, and management of long-haul trucking operations using GPS, sensors, and mobile connectivity. 5. To implement intelligent route optimization and capacity utilization algorithms that minimize transit time, fuel consumption, and empty return trips while maximizing efficiency and profitability. 6. To develop a responsive alert and communication module that ensures timely updates on delays, maintenance needs, or emergencies, enabling informed decision-making and improved safety across the transport network.
20	**Mr. Anesh Ashok Kshirsagar**	**Problem 1:** Smart Attendance System using Face Recognition **Objectives:** 1. Implement real-time face detection and recognition using a webcam and open-source libraries like OpenCV and face-recognition. 2. Implement real-time face detection and recognition using a webcam and open-source libraries like OpenCV and face-recognition. 3. Develop a user-friendly dashboard for instructors to view attendance records, generate reports, and manage student data.

		Problem 2: AI-Powered Chatbot for College Enquiries **Objectives:** 1. Develop a conversational chatbot using NLP techniques and frameworks like Dialog flow, Rasa, or custom models with transformers. 2. Integrate the chatbot into a web interface (or mobile app) for easy access by students and visitors 3. Enable continuous learning and feedback loop so the chatbot can improve its accuracy and handle new queries over time. **Problem 3:** Intelligent Timetable Management System for Engineering Faculty **Objectives:** 1. Implement algorithms to automatically detect scheduling conflicts such as overlapping classes for faculty, classrooms, or students, and alert the user in real-time. 2. Design a scheduling assistant that suggests optimal available time slots based on faculty availability, classroom resources, and course requirements to prevent manual trial-and-error. 3. Provide a user-friendly web-based interface for adding, editing, and viewing timetables, with drag-and-drop functionality and export options (PDF/Excel) for easy sharing.
21	Ajay B Kapase	**Problem 1:** AI based Question paper evaluation system **Objectives:** 1. Automatically grade student responses and generate detailed feedback based on predefined evaluation criteria. 2. Use NLP algorithms to assess the quality, relevance, and coherence of written responses to subjective questions. 3. Design a scalable AI system that can handle large-scale assessments, providing real-time evaluations and reducing grading time. **Problem 2:** Abnormal Event Detection & Reporting system **Objectives:** 1. Continuously monitor data streams (e.g., video, sensor inputs) to detect unusual or anomalous events using machine learning models or statistical analysis.

		2. Automatically classify detected anomalies, categorize their severity, and trigger real-time alerts to relevant stakeholders (e.g., security, operations teams). 3. Provide an intuitive dashboard to visualize detected abnormal events, offer detailed reports, and enable easy access to historical data for trend analysis and decision-making. **Problem 3:** "Multi-Modal Deepfake and Image Forgery Detection System **Objectives:** 1. Develop a robust visual anomaly detection engine for images and video frames. 2. Integrate an audio analysis module for synthetic speech and lip-sync inconsistencies. 3. Construct a multi-modal fusion framework for comprehensive authenticity verification.
22	Mr. Uday Dhanaji Patil	**Problem 1:** RISK AND OPPORTUNITY PREDICTION FOR INDUSTRIAL APPLICATION USING DATA AUGMENTATION FOR SENTIMENT ANALYSIS **Objectives:** 1. The prediction of risks and opportunities is essential for industrial applications to make informed decisions and mitigate potential negative consequences while capitalizing on positive outcomes 2. Sentiment analysis has emerged as a valuable tool in predicting risks and opportunities by analyzing the sentiment expressed in text data such as customer reviews, social media posts, and news articles. 3. We present a framework that involves collecting and preprocessing text data from various sources, augmenting the dataset using techniques such as back-translation and contextual word replacement, and training a sentiment analysis model using machine learning algorithms. **Problem 2:** Suraksha: Enhancing Safety and Emergency Response **Objectives:** 1. This Project monitor vehicles real time location and in the event of accident, system will detect , track the vehicle. 2. Suraksha is the revolutionary mobile application, designed to provide authorized user with instant access to critical emergency by simply scanning QR code. 3. This is IOT Based device will provide necessary information in natural calamities, accident, theft situation.

		Problem 3: Smart Irrigation System Using IoT. **Objectives:** 1. To develop an automated irrigation system that monitors soil moisture and weather conditions in real-time 2. To optimize water usage in agriculture by irrigating only when necessary. 3. To remotely control and monitor irrigation through a mobile or web application.
23	Avinash Powar	**Problem 1:** AI and Generative AI-Driven Automation for Multi-Cloud and Hybrid Cloud Architectures: Enhancing Security, Performance, and Operational Efficiency **Objectives:** 1. Proactive Threat Detection and Prevention. 2. Automated Incident Response. 3. Intelligent Workload Placement and Optimization **Problem 2:** Cloud Workload Turning Points Prediction via Cloud Feature-Enhanced Deep Learning **Objectives:** 1. To develop smart waste management system Data Collection and Feature Engineering 2. Deep Learning Model Development and Optimization 3. Turning Point Detection and Evaluation **Problem 3:** Smart Edge: Smart Healthcare End-to-End Integrated Edge and Cloud Computing System for Diabetes Prediction Enabled by Ensemble Machine Learning **Objectives:** 1. Edge Layer Development and Optimization Effective AI Based solution 2. Cloud Layer Development and Ensemble Machine Learning Integration 3. System Integration, Performance, and Security and Clinical Impact and User Experience
24	Mrs. Yogita Sachin Narule	**Problem 1:** Implementing a scalable system for monitoring and managing mental health in large groups, such as workplaces and families. **Objectives:** 1. Early Detection of Mental Health Issues. 2. Privacy-Preserving Data Collection 3. Personalized Mental Health Insights 4. **Problem 2:** Creating an affordable health tracking system to promote regular medical check-ups, especially in rural areas.

		Objectives: 1. Improve Access to Preventive Healthcare 2. Track Essential Health Metrics 3. Integration with Local Healthcare Infrastructure **Problem 3:** Developing an AI-driven solution for quick, accurate, and effortless patient record analysis and monitoring. **Objectives:** 1. Automate Patient Record ProcessingEffective AI Based solution 2. Multilingual and Context-Aware Functionality 3. Protect Patient Privacy and Data Security